

PEDRO BARBOSA

Backend Software Engineer — Python, Go, PHP, SQL

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Portuguese (native) · English (fluent) · German (basic)

SUMMARY

Backend engineer building and maintaining software that runs in production for real users. I write clean, modular code, work from clear data structures and relational modelling, and back every change with automated tests. Day to day I debug live systems, find root causes, resolve defects and optimise slow queries and requests. I work in Agile teams, review code, and use Git on every project.

EXPERIENCE

Full Stack Developer — TIIX (remote)

April 2026 – present

- One of two developers on a healthcare platform used daily by patients, clinics and professionals.
- Design and build backend features end to end: relational schema and versioned migrations, REST endpoints, and the automated tests that protect them.
- Troubleshoot and resolve production defects and performance issues: reproduce the failure, find the root cause, fix it, and add a regression test.
- Wrote 23 of the repository's 50 database migrations and the largest share of its automated test suite.
- Take part in code reviews and in technical decisions with the team.

Software Engineer — QuantConnect (freelance, remote)

September 2025 – present

- Python services and pipelines on the LEAN engine: backtesting over large time-series datasets, research models and risk metrics.
- Focus on correctness and reproducibility: exact numerical handling, deterministic results, batch runs over years of data.

Python Developer — Freelance via Workana (remote)

January 2025 – March 2026

- Django system for a non-profit: role-based access control, email verification, privacy-compliant handling of personal data, normalised relational schema built to stay fast as data grows.
- Data pipelines with pandas and openpyxl replacing manual spreadsheet processes, removing a recurring source of human error.
- Dashboards over fleet, HR and inventory data for decision making.

Software Development Intern — TT&T Soluções em Informática (remote)

August 2024 – January 2025

- Django backend: orders, authentication, delivery tracking and payment gateway integration.
- MySQL modelling and documentation, unit tests with pytest, optimisation of slow queries and HTTP requests.

SELECTED WORK

iode — open source daemon in Go with SQLite: indexes projects through a versioned subprocess contract, with bounded timeouts and a job queue with backoff and orphan recovery. 5,100 lines of code covered by 5,675 lines of tests.

Akaunting — merged contribution to an open source accounting platform: fixed database host and port not being applied to the connection during install.

SKILLS

Languages: Python, Go, PHP, JavaScript, TypeScript, SQL, Bash

Frameworks: Django, Laravel, React, pandas, scikit-learn

Databases: MySQL, PostgreSQL, SQLite — modelling, queries, versioned migrations

Tooling: Git, Docker, GitHub Actions, GitLab CI, Linux, systemd

Practices: Agile/Scrum, code review, automated testing (unit and integration), REST API design, debugging and performance optimisation

EDUCATION AND CERTIFICATIONS

Computer Engineering (Data and Machine Learning) — Centro Universitário IBMR · 2023 – 2028 (expected) · GPA 8.7/10

Scientific initiation in Economic Mathematics and Machine Learning certificate — IMPA, 2024

HackerRank: Problem Solving (Intermediate), REST API (Intermediate), Python · Coursera: Python for Data Science and AI